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Digital Passion

Fuelling The Pursuit Of Technology Knowledge

Tomorrow

TOP TEN Technologies TO LOOK OUT FOR

Madhusudan Mukerjee

From a few favourite songs on magnetic tape on a Walkman, to wireless, portable MP4 movies; from beepers to cell phones; from SLRs to camera phones – in just a couple of decades, science has taken us beyond the predictions of futurology and into the realms of Asimov and Arthur C Clarke. In such an intensely fluxed technological environment, we've become so used to witnessing Spidey-like jumps in technology during our lifetimes that even touch screens are beginning to seem old hat. We at *Digit* share your impatience and so we decided to satisfy our curiosity – and yours – by ferreting out and laying before you ten of the most remarkable technologies being worked on today, which are set to bring sci-fi to reality.

1. CUBIC CHIPS

Laying It On Thick

Stuck In The Moore

In 1965, Intel's Gordon Moore stated what has come to be known as Moore's Law – that the number of transistors on a chip will double about every two years (how many times have you heard that one before?). But as chips get smaller, engineers are already facing problems in trying to cram innumerable transistors into decreasing space.

The Rochester Chip

Enter the Rochester chip – a chip that's been designed vertically, bottom up, specifically to maximise the main functions of a chip by the use of several layers of processors. However this '3D' chip is unlike the 'stacking' idea – where present-day chips are merely stacked one on top of another. This one is built so that each layer interacts with each other layer as a part of a single circuit, while performing different functions. Chips for audio, for example, differ in requirements from chips that process digital photos or texts. The Rochester chip is designed simultaneously to deal with the different speeds and power requirements of these processes.